



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Department of Computer Science
Faculty of Science & Technology (FST)**

Fall 2024-25

Section: H, Group: 01

PROJECT TITLE

**Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online
Learning Environment**

Supervised By TONNY SHEKHA KAR

Submitted By

Group Members

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Test case:

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AFNAN BIN ISLAM NAHIN		
Test Case ID: VC_01		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: AFNAN		
Module Name: Login Session		Test Execution date:01-1-2025		
Test Title: Verify login with valid email and password				
Description: Test the login functionality for valid credentials				
Precondition (If any): User must have a valid account				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website. 2. Enter email and password. 3. Click the "Continue" button.	URL: www.vclassroom.com Email: afnan@gmail.com Password: afnan123	Website loads successfully. User credentials are entered. User is logged In and redirected to the dashboard.	As expected	Pass
Post Condition: User successfully logs in and accesses the dashboard.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AFNAN BIN ISLAM NAHIN		
Test Case ID: VC_12		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: AFNAN		
Module Name: User Registration (Sign-Up)		Test Execution date:01-1-2025		
Test Title: Test system validation for invalid data and duplicate accounts				
Description: Test the functionality of creating a new user account				
Precondition (If any): User must not have an existing account with the same email or username.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Sign-Up page. 2. Enter valid user details. 3. Confirm password. 4. Click the "Sign Up" button.	URL: www.vclassroom.com/signup Full Name: AFNAN NAHIN Age : 18 Email: afnan@gmail.com Password: afnan123 Confirm Password: afnan123	Sign-Up page is displayed. User details are entered successfully. Password matches. User is registered successfully and redirected to the dashboard.	As expected	Pass
Post Condition: A new user account is created, and the user is logged into the platform.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AFNAN BIN ISLAM NAHIN		
Test Case ID: VC_13		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: AFNAN		
Module Name: Notifications		Test Execution date:01-1-2025		
Test Title: Verify users receive push notifications				
Description: Test the push notification system for events and updates				
Precondition (If any): Notifications must be enabled in user settings.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Schedule a meeting. 2. Publish a new assignment.	Meeting ID: 50505 Assignment_Title: Homework 1	Notification for scheduled meeting is sent. Notification is sent to your friend.	As expected	Pass
Post Condition: Notifications are sent and received as expected.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AMAN ULLAH		
Test Case ID: VC_20		Test Designed date: 27-12-2024		
Test Priority (Low, Medium, High): Medium		Test Executed by: AMAN		
Module Name: Attendance Tracking		Test Execution date:02-1-2025		
Test Title: Verify attendance tracking during a session				
Description: Test automatic attendance marking for participants				
Precondition (If any): Attendance tracking is enabled for the session.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start a live session. 2. End the session.	Meeting ID: 50505	Attendance is recorded for all participants who join. Attendance report is generated.	As expected	Pass
Post Condition: Attendance details are saved and visible in the reports.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AMAN ULLAH		
Test Case ID: VC_25		Test Designed date: 27-12-2024		
Test Priority (Low, Medium, High): Low		Test Executed by: AMAN		
Module Name: Session Recording		Test Execution date:02-1-2025		
Test Title: Verify session recording functionality				
Description: Test recording and playback of live sessions.				
Precondition (If any): Recording option must be enabled				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start recording during session. 2. Stop recording.	Meeting ID: 70707	Session recording starts successfully. Recording is saved and accessible for playback.	As expected	Pass
Post Condition: Recording is stored and available for review.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: HRITTIK BHATTACHARJEE		
Test Case ID: VC_26			Test Designed date: 28-12-2024		
Test Priority (Low, Medium, High): Medium			Test Executed by: HRITTIK		
Module Name: Screen Sharing			Test Execution date:03-1-2025		
Test Title: Verify instructor can share the screen					
Description: Test screen-sharing functionality during a session					
Precondition (If any): The instructor must be in a live session					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start a live session. 2. Click on "Share Screen."		Meeting ID: 70707 No additional data	Live session starts successfully. The screen is successfully shared with all participants.	As expected	Pass
Post Condition: Participants can view the instructor's screen without interruptions.					

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: HRITTIK BHATTACHARJEE	
Test Case ID: VC_27			Test Designed date: 28-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by: HRITTIK	
Module Name: Join Meeting			Test Execution date:03-1-2025	
Test Title: Verify user can join a live session				
Description: Test the functionality of joining a live session.				
Precondition (If any): The user must have a valid account and a scheduled session.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in to the platform. 2. Navigate to the session. 3. Click "Join Meeting."	Meeting ID: 70707	The user successfully logs in. The session details are visible. User successfully joins the session.	As expected	Pass
Post Condition: User is connected to the session, and audio/video functionality works seamlessly.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: MD.HACHIBUR RAHMAN		
Test Case ID: VC_28		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: HACHIBUR		
Module Name: Create Meeting		Test Execution date:01-1-2025		
Test Title: Verify a meeting can be created with valid details				
Description: Test the functionality of scheduling a new meeting.				
Precondition (If any): The user must be logged in to the platform.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the "Create Meeting" page. 2. Enter meeting details. 3. Click the "Create Meeting" button. 4. Add participants.	URL: www.vclassroom.com/create-meeting Participant_Emails: user1@example.com, user2@example.com Meeting Title: Team_Discussion Date:31/12/2024 Time:10:00am Duration: 60 mins	Create Meeting page is displayed. Meeting details are entered successfully. Participants are added to the meeting. Meeting is successfully created, and meeting ID is generated.	As expected	Pass
Post Condition: A new meeting is created, and participants receive an invitation with the meeting details (ID and passcode).				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: MD.HACHIBUR RAHMAN	
Test Case ID: VC_30			Test Designed date: 25-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by: HACHIBUR	
Module Name: Chat Feature			Test Execution date:01-1-2025	
Test Title: Verify sending a chat message				
Description: Test the ability of a user to send a chat message in a meeting.				
Precondition (If any): User must be logged in and in the chat window.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the chat window. 2. Leave the message input blank. 3. Click the "Send" button.	Message: "Hello, everyone!"	Chat window is displayed. The message is visible in the input box. Message is sent and appears in the chat.	As expected	Pass
Post Condition: Messages from other participants are displayed in the chat window in real-time.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: AFNAN BIN ISLAM NAHIN	
Test Case ID: VC_32			Test Designed date: 25-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by: AFNAN	
Module Name: Call Feature			Test Execution date:01-1-2025	
Test Title: Verify starting a call				
Description: Test if a user can successfully start a video or audio call.				
Precondition (If any): User must be logged in and part of a virtual classroom.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the "Call" section. 2. Click the "Start Call" button.	Call Type: Video/Audio	Call interface is displayed. Call initiates, and "Call in Progress" is shown.	As expected	Pass
Post Condition: The user is in an active call session.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: AFNAN BIN ISLAM NAHIN	
Test Case ID: VC_33			Test Designed date: 25-12-2024	
Test Priority (Low, Medium, High): Medium			Test Executed by: AFNAN	
Module Name: Create Assignment			Test Execution date:01-1-2025	
Test Title: Verify creating an assignment				
Description: Test if a user can successfully create an assignment.				
Precondition (If any): User must be logged in and have access to the assignment creation module.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the "Assignment" section. 2. Click on the "Create Assignment" button. 3. Enter assignment details. 4. Attach resources . 5. Click on "Share" button.	Title: Math Assignment Description: Solve problems 1-10 Due Date: 2025-01-10 File: "Problems.pdf"	Assignment creation page is displayed. Assignment creation form is displayed. Assignment details are entered successfully. File is successfully uploaded and attached to the assignment. Assignment is successfully created and share.	As expected	Pass
Post Condition: The assignment is visible to those the user share it with .				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: AFNAN BIN ISLAM NAHIN	
Test Case ID: VC_50			Test Designed date: 25-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by: AFNAN	
Module Name: Create Exam			Test Execution date:01-1-2025	
Test Title: Verify creating an exam				
Description: Test if a user can successfully create an exam.				
Precondition (If any): User must be logged in and have access to the exam creation module.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the "Assignment" section. 2. Click on the "Create Exam" button. 3. Enter exam details. 4. Add questions. 5. Click on "Share" button.	Title: Final Math Exam Description: Covers Chapters 1-5 Duration: 90 minutes Time: 10 AM Questions: Multiple Choice, Short Answers, True/False	Exam creation page is displayed. Exam creation form is displayed. Exam details are entered successfully. Questions are added to the exam successfully. Exam is successfully created and scheduled.	As expected	Pass
Post Condition: The exam is visible to those the user share it with and is scheduled for the specified date and time.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment			Test Designed by: WALID BIN KHALID	
Test Case ID: VC_51			Test Designed date: 25-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by: WALID	
Module Name: Attend Exam			Test Execution date:01-1-2025	
Test Title: Verify the ability to start an exam				
Description: Test if a user can successfully start an exam				
Precondition (If any): A published exam must exist, and the user should be enrolled in the respective course.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Navigate to the "Assienment" section. 2.Click the "Attend Exam" button. 3. Select an exam.	Exam Title: Final Math Exam	List of available exams is displayed. Exam details and "view Exam" button are displayed. The exam interface is displayed, and the timer starts.	As expected	Pass
Post Condition: The user successfully starts the exam and can see the questions along with the timer.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AFNAN BIN ISLAM NAHIN		
Test Case ID: VC_52		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: AFNAN		
Module Name: Submit Assignment		Test Execution date:01-1-2025		
Test Title: Verify successful assignment submission				
Description: Test if a user can upload and submit an assignment successfully.				
Precondition (If any): The assignment must be published by a user, and to those the user share it should be enrolled in the respective course.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the "Assignments" section. 2.Click on "Submit Assignment." 3. Select the assignment. 4. Click on "Attach Assignment." 5. Click on "Turn in."	Exam Title: Final Math Exam File: math_homework1.pdf	. Student dashboard is displayed. List of available assignments is displayed. Assignment details page is displayed. File is uploaded successfully.	As expected	Pass
Post Condition: The assignment is submitted, and the submission timestamp is recorded in the system.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: WALID BIN KHALID		
Test Case ID: VC_53		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: WALID		
Module Name: Calendar		Test Execution date:01-1-2025		
Test Title: Verify event creation in the calendar				
Description: Test if users can create and add events to the calendar.				
Precondition (If any): User must be logged in and have access to the calendar module.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in to the platform. 2. Navigate to the "Calendar" section. 3. Click on "Add Event."	Event Title: "Math Quiz" Start Date: 2025-01-10 End Date: 2025-01-10 Time: 10:00 AM	. Calendar interface is displayed. Event is added to the calendar with the correct details.	As expected	Pass
Post Condition: The event is saved and displayed on the calendar.				

Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment		Test Designed by: AFNAN BIN ISLAM NAHIN		
Test Case ID: ST_01		Test Designed date: 25-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by: AFNAN		
Module Name: Settings Page		Test Execution date:01-1-2025		
Test Title: Verify enabling and disabling notification preferences				
Description: Test the functionality of toggling notification preferences in the Settings Page.				
Precondition (If any): • The updated settings should persist after refreshing the page. • Notifications should function according to the enabled or disabled preferences.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Navigate to the Settings Page. 2.Toggle "Message Reminders" to ON . 3Toggle "Files and Links Share" to OFF . 4. Toggle "End-to-end encrypted calls" to ON . 5. Select a different language from the dropdown menu.	Select "English (UK)"	Settings page loads successfully. The toggle should switch to ON . The toggle should switch to OFF . The toggle should switch to ON . Language preference should update successfully.	As expected	Pass
Post Condition: • The updated settings should persist after refreshing the page. • Notifications should function according to the enabled or disabled preferences.				